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ORIGINAL PAPER

NCA-EVA: An Innovative Ensemble-Based Approach for Alzheimer's Disease Detection from Magnetic Resonance Imaging

Esra Yüzgeç Özdemir[1,2] · Canan Koç[1] · Fatih Özyurt[1]

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Abstract
Alzheimer's disease is a progressive neurodegenerative disorder that is challenging to diagnose at an early stage. Affecting over 55 million people worldwide, its prevalence is expected to rise sharply by 2030. The use of artificial intelligence (AI) techniques has become increasingly important to improve the speed and accuracy of diagnosis. In this study, we propose the NCA-Enhanced Voting Algorithm for Alzheimer's Classification (NCA-EVA) to support computer-aided diagnosis. A total of 66 models were trained for four-class data and six models for two-class data. The proposed method successfully classified all four stages of Alzheimer's disease, achieving 98.97% accuracy in four-class classification and 99.87% accuracy in binary classification. Moreover, with a processing time of just 1.26 s, NCA-EVA is approximately 1200 times faster than a comparable study using NCA-based feature selection. These findings demonstrate that Alzheimer's diagnosis can be performed both quickly and with high accuracy, and the proposed approach has potential applications in other healthcare data analysis tasks.

Keywords Alzheimer's disease · Artificial intelligence · Vision Transformers · CNN · Feature extraction

Introduction

Background

Alzheimer's disease (AD) is a neurodegenerative disorder that results in cognitive decline in the brain. This disease causes the decline of cognitive functions and progressively worsens over time. This trend results in a decline in cognitive function throughout the disease [1]. Alzheimer's disease is characterized as a progressive condition with five primary stages [2]. This disease is challenging to identify in everyday life as it does not exhibit conspicuous signs until it reaches severe stages. If not recognized promptly, the condition will advance and cause harm to the brain.

Currently, there is no medication available to halt or reverse Alzheimer's disease. The therapy approach does not offer a cure for the condition, and the timing and causes of this chronic degenerative process remain mostly unknown. Diagnosis of this condition is challenging until severe symptoms are evident, and it is believed to be influenced by environmental or genetic factors, including previous head trauma, hypertension, and depression [3]. The 2021 Alzheimer's Disease International (ADI) report indicates that a survey was administered to 1111 multidisciplinary doctors, 205 dementia patients, and 2122 dementia caregivers. This poll indicates that 47% refrain from seeking a diagnosis due to insufficient access to qualified specialists, 34% due to the financial burden of diagnosis and treatment, and 46% due to apprehension about a dementia diagnosis. These rates have been recognized as substantial impediments to diagnosis. Additional obstacles to clinical diagnoses encompass restricted access to specialist diagnostic tests (38%) and the perception among patients and their families that no interventions are possible following a dementia diagnosis (33%) [4]. Given these percentages, the early diagnostic rate of Alzheimer's disease is deemed highly challenging.

Globally, 55 million individuals have received a diagnosis of Alzheimer's disease. This figure is anticipated to rise dramatically by 2030. Given the clinical, social, and economic circumstances, early diagnosis is significantly challenging. Given the numerous illness outbreaks globally and the prevalent skepticism towards treatment protocols, the rising incidence of Alzheimer's disease is accompanied by challenges

✉ Fatih Özyurt
ozyurtfatih@gmail.com

[1] Software Engineering, Engineering Faculty, Firat University, Elazığ, Turkey

[2] Computer Engineering, Engineering and Architecture Faculty, Bingol University, Bingöl, Turkey

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in diagnostic and therapeutic intervention. Furthermore, the ADI report released in 2021 indicated that almost 30% of fatalities ascribed to Alzheimer's disease were inaccurately diagnosed. Re-evaluations with computerized imaging scans were performed on these patients, yielding a 25% correction in diagnosis. Given this situation, the utilization of magnetic resonance imaging (MRI) and computed tomography (CT) scans, which are readily available technologies, in conjunction with clinical assessments for diagnosing dementia and Alzheimer's disease, is a significant measure in mitigating potential diagnostic errors.

Alzheimer's disease is a disorder that impairs cognitive skills due to neurodegenerative processes. The aggregation of beta-amyloid plaques and tau protein tangles constitutes the basic foundation of the disease. These aggregates impede neural communication, causing synapse loss and ultimately culminating in neurodegeneration marked by neuronal death [5]. These biological processes predominantly influence brain regions associated with memory and cognitive functioning, including the hippocampus and cortex. The demise of neuronal cells and synaptic degradation results in significant deficits in memory, language, problem-solving, and daily functioning.

The initial phases of Alzheimer's disease are characterized by symptom ambiguity and their conflation with aging indications, which complicates diagnosis. Early diagnosis is essential for the prompt commencement of therapies designed to mitigate disease development [6]. Nonetheless, the existing diagnostic approaches lack adequate precision at this stage. Imaging modalities like MRI and CT scans are essential instruments for identifying structural alterations in the brain; however, these changes generally manifest in the advanced stages of the disease. Advanced technologies, including biomarker analysis and PET scans for early diagnosis, provide enhanced accuracy; however, their extensive clinical application is restricted by cost and accessibility limitations [7].

AI-assisted diagnostic methods represent a significant advancement for the early detection of Alzheimer's disease. AI systems can scrutinize extensive datasets to identify nuanced alterations in brain pictures, facilitating the early identification of the condition. These devices facilitate the expedited and precise assessment of MRI and CT imaging outcomes, specifically [8]. AI-driven decision support systems minimize inaccuracies in clinical diagnostic procedures and enhance treatment administration. AI-based decision support systems enhance treatment management and lower errors in clinical diagnosis procedures. The growing application of AI in healthcare has novel prospects for the early diagnosis and therapy of neurodegenerative disorders such as Alzheimer's [9].

In the advanced stages of Alzheimer's disease, alterations in brain structure become increasingly evident, but in the initial stages, these changes are sometimes imperceptible or indistinguishable. Consequently, in addition to conventional imaging techniques, sophisticated computer-aided diagnostic technologies are essential for the early detection of the disease [10]. These systems facilitating early diagnosis seek to identify the disease at phases where biological alterations have not yet manifested as clinical symptoms through the analysis of brain imaging data. This circumstance is anticipated to provide favorable results, including enhanced early intervention chances and an improvement in the patient's quality of life. Artificial intelligence and other technological advancements significantly enhance the management of complicated neurological disorders, such as Alzheimer's disease, by improving the efficacy and accessibility of healthcare services [11].

Motivation

Deep learning architectures and multimodal data fusion approaches have significantly propelled recent improvements in the diagnosis of Alzheimer's. Numerous studies in the literature indicate that these methods provide substantial advantages in enhancing diagnostic precision and streamlining processing durations. Prasath and Sumathi devised a deep learning technique utilizing the Pipelined LeNet (PLN) architecture for the early detection of Alzheimer's disease. This approach employed an image fusion technique to improve the interior areas of low-resolution MRI images. It differentiated between Alzheimer's and healthy brain images with 99.5% accuracy utilizing a PLN architecture including six convolutional layers and six sampling layers [12].

Qiao et al. have created a deep learning model named Convolutional-Squeeze-Excitation-Softmax-NET (CSES-NET). This model categorizes Alzheimer's disease, early mild cognitive impairment (EMCI), late mild cognitive impairment (LMCI), and healthy controls by synthesizing multi-channel data, including voxel-based morphometry, surface morphometry, and radiometric characteristics derived from MRI data. The CSES-NET model has exhibited exceptional performance, achieving an accuracy rate of 96.17%, and has analyzed the correlations between genetic and radiomic aspects of Alzheimer's disease [13].

Lu et al. have implemented a methodology that amalgamates MRI, genetic, and clinical data for Alzheimer's diagnosis through the creation of the Hierarchical Attention-Based Multi-Modal Fusion (HAMF) model. This model has been highly effective in assessing the risk of progression from mild cognitive impairment (MCI) to Alzheimer's disease and has emerged as a viable instrument for the early diagnosis of Alzheimer's, with an accuracy rate of 87.2% [14].

Sasikala et al. created a CAD system that uses MRI data to diagnose Alzheimer's disease in their published papers from 2024. Structures like gray matter (GM), white matter

(WM), and cerebrospinal fluid (CSF) were correctly identified during brain tissue segmentation using the iteratively reweighted adaptive spatial fuzzy c-means (IRW-ASFCM) approach. Features were extracted using a 2LS-WaveCNN ensemble, which is a two-level selective wavelet kernel convolutional neural network. After that, the neural network layers were trained using the deep tree training (DTT) method. The specificity rate was 99.64%; the accuracy rate was 99.26%, and the precision rate was 99.26% among the Alzheimer's, cognitively normal (CN), and MCI classes, respectively. The investigation of the proposed model's processing time also showed that it outperforms previous models in terms of performance and time complexity; the model took 97.88 s to finish the whole processing [15].

In their 2024 paper, Zhao et al. performed an exhaustive analysis of AI applications for Alzheimer's disease. This study addresses the application of AI in etiology discovery, computer-aided diagnostic (CAD), computer-aided prognosis (CAP), and treatment domains. The objective has been to enhance the diagnosis and prognosis of Alzheimer's disease by the utilization of multi-modal data, including genetic, neuroimaging, and linguistic information. The research has attained application rates of 65% in CAD systems, 11% in prognostic studies, 19% in biomarker identification, and 5% in treatment utilizing AI. Zhao and his associates asserted that AI methodologies may enhance clinical decision-making by offering efficient solutions for disease detection and therapy [16].

In a 2024 study, Sorour et al. introduced a deep learning (DL) methodology utilizing MRI data for the early diagnosis of AD. The study employed five distinct deep learning models: CNNs without augmentation, CNNs with augmentation, CNNs combined with LSTM and augmentation, CNNs integrated with SVM and augmentation, and VGG16 combined with SVM and augmentation. These models were autonomously trained for feature extraction and classification tasks. The models are categorized based on the application of data augmentation, with particular emphasis on the VGG16-SVM model, which is enhanced by transfer learning techniques. The testing findings demonstrated that the CNN-LSTM model achieved an accuracy of 99.92%, indicating improved performance [17].

Huang et al. devised a novel approach in their 2024 research, integrating the Deep Forest algorithm with near-infrared (NIR) spectroscopy for the diagnosis of AD and vascular dementia (VaD). This study evaluated serum samples from the AD and VaD groups utilizing the aquaphotomics technique, which investigates the influence of water molecules on hydrogen bonding. The Deep Forest model has surpassed typical machine learning techniques, attaining an accuracy of 98.67%, sensitivity of 97.33%, and specificity of 100%. This approach integrates the rapidity and precision of NIR spectroscopy with the robust modeling proficiency of deep learning, facilitating the swift and accurate distinction between AD and VaD [18].

Rezaee and Zhu, in their 2024 work, devised a deep learning model that combines CascadeNet with manually constructed characteristics to identify Alzheimer's disease (AD) and mild cognitive impairment (MCI) from EEG waves. In the study, EEG signals were initially divided into subbands by discrete wavelet transform (DWT), and then, data augmentation was executed via the windowing method. The enhanced CascadeNet model was subsequently employed for feature extraction and classification tasks involving these signals. This model has demonstrated efficacy in detecting Alzheimer's disease (AD) and mild cognitive impairment (MCI), particularly with limited datasets, by attaining high accuracy in spectral pictures derived from EEG signals. The fine-tuning and training duration for each CNN block of the Cascade-ResNet-18 model is designated as 4 to 6 h. The research attained accuracy rates of 98.84% and 97.78% on the Figshare and Brazil datasets, respectively, indicating enhanced performance relative to current methodologies [19].

Ayus and Gupta devised a hybrid model named Hybrid Recurrent Ensemble Network (HReENet), employing deep learning methodologies for the diagnosis of Alzheimer's disease in their 2024 article. This model seeks to categorize the four stages of Alzheimer's disease (NonDemented, VeryMildlyDemented, MildlyDemented, ModeratelyDemented) by integrating various deep learning architectures, including CNN, Conv1D-LSTM, and LSTM. The model utilized the Kaggle Alzheimer's classification dataset, attaining an accuracy of 99.97%, a precision of 99.98%, a recall of 99.95%, and an F1-score of 99.97%. The suggested model has exhibited enhanced performance relative to previous models, including CNN and LSTM, markedly improving classification accuracy [20].

Rani et al. developed an optimal clustering-based pre-trained SqueezeNet model for the early diagnosis of Alzheimer's disease in their study. The research employed the Adaptive Reptile Search Algorithm (ARSA) for the segmentation of gray matter (GM), white matter (WM), and cerebrospinal fluid (CSF). ARSA enhances segmentation accuracy by refining the identification of cluster centers. The pre-trained SqueezeNet model was subsequently employed to classify various phases of Alzheimer's disease. The experimental studies of the ADNI and OASIS datasets indicated that the model attained accuracy rates over 98%. The results demonstrate that the model provides an efficient approach for diagnosing Alzheimer's disease [21].

Alp et al. developed a method that combines Vision Transformer (ViT) and Time Series Transformer (TST) architectures to identify Alzheimer's disease (AD) [22]. To solve the problems encountered during the processing of 3D MRI data, the images were divided into two different

three-dimensional slices, and transfer learning techniques were used. The ADNI dataset was used for binary and multiclass classifications. The results showed that the method demonstrated superior performance with 99.048% accuracy in binary classification and 99.014% accuracy in multiclass classification. These findings highlight the effectiveness of sequence modeling methods and self-attention mechanisms in medical image analysis, providing a powerful solution for the early detection and classification of Alzheimer's disease. Literature summary of the reviewed studies is given in Table 1.

Recent studies have shown that deep learning-based approaches achieve very high accuracy rates in the diagnosis of Alzheimer's disease [12–22]. However, these methods have significant limitations. Many studies focus solely on binary classification and are insufficient in simultaneously detecting the different stages of Alzheimer's disease (No Dementia, Very Mild Dementia, Mild Dementia, Moderate Dementia) with high accuracy. In addition, many models in the literature have a long processing time, making it difficult to use them as real-time decision support systems in clinical settings. Some methods focus only on a single data modality (e.g., MRI only), neglecting the combined use of different feature extraction techniques and dimension reduction methods. Considering these shortcomings, there is a need for a new method that enables early and accurate diagnosis of Alzheimer's disease, while also offering high performance on multiclass data and operating quickly. Motivated by this, the NCA-EVA method proposed in our study offers significant improvements in both accuracy and processing time compared to existing approaches in the literature for multistage Alzheimer's classification.

Table 1 Related works

Year	Authors	Study	Dataset	Accuracy	Execution time (s)
2024 [12]	Prasath and Sumathi	Pipelined LeNet (PLN)	Kaggle Alzheimer's classification dataset (KACD) (2 classes)	99.5%	0.65
2024 [13]	Jianping Qiao et al	Convolutional-Squeeze-Excitation-Softmax-NET (CSES-NET)	ADNI (2 classes)	96.17%	Not specified
2024 [14]	Peixin Lu et al	Hierarchical Attention-Based Multimodal Fusion (HAMF) Model	ADNI (2 classes)	87.2%	Not specified
2024 [15]	T.S. Sasikala	IRW-ASFCM and 2LS-WaveCNN-based CAD system	Custom collected dataset (3 classes)	99.54%	97.88
2024 [16]	Qing Zhao et al	CAD systems	Multi-modality (genetics, neuroimaging) (3 classes)	65% (CAD)	Not specified
2024 [17]	Sorour et al	CNN-LSTM-With-Aug	ADNI (2 classes)	99.92%	Not specified
2024 [18]	Wenchang Huang et al	Deep Forest	Serum samples (NIR spectroscopy) (3 classes)	98.67%	Not specified
2024 [19]	Khosro Rezaee and Min Zhu	CascadeNet	Figshare, Brazilian EEG (3 classes)	98.84%, 97.78%	Not specified
2024 [20]	Ishan Ayus and Deepak Gupta	CNN-Conv1D-LSTM and HReENet (4 classes)	Kaggle Alzheimer's classification dataset (KACD)	99.97%	Not specified
2025 [21]	K. Emily Esther Rani and S. Baulkani	ARSA-CA + pre-trained SqueezeNet (4 classes)	ADNI	98.4%	Not specified
2024 [22]	Alp et al	Vision Transformer (ViT) + Time Series Transformer (TST) (2 classes–4 classes)	ADNI (2 classes)	99.048%	Not specified
			ADNI (4 classes)	99.014%	Not specified
Proposed method		NCA-EVA	2 classes	99.87%	0.74
			4 classes	98.97%	1.26

Material and Method

Dataset

Alzheimer's disease is a neurological disorder that is becoming more prevalent due to the aging global population, resulting in significant repercussions such as memory impairment and cognitive decline. The accurate diagnosis of the disease and its stage classification are crucial for comprehending its progression and for implementing suitable therapy and intervention strategies. Consequently, brain images acquired by magnetic resonance imaging (MRI) techniques serve as a significant data source for identifying various phases of Alzheimer's disease. The dataset included in this research comprises images depicting four distinct stages of Alzheimer's disease, systematically classified and balanced to categorize the disease's symptoms [23]. The dataset was obtained from the publicly available Kaggle platform, where it is shared under an open data usage license. Therefore, no ethical approval was required for its use in this study.

The Alzheimer's multi-class dataset is an open-source collection of MRI scans depicting various phases of Alzheimer's disease for diagnostic and classification purposes. This multi-class dataset comprises around 44,000 MRI scans, classified into four distinct categories based on the severity of Alzheimer's disease. This dataset, designed for training and evaluating machine learning models, comprises MRI images exclusively featuring brain tissue devoid of any non-cranial tissue components. Sample images of Alzheimer's stages contained in the dataset are shown in Fig. 1.

The dataset comprises four primary files, each corresponding to a distinct severity class of Alzheimer's disease. Each folder comprises images corresponding to a particular stage of the disease: NonDemented (12,800 data), VeryMildDemented (11,200 data), MildDemented (10,000 data), and ModerateDemented (10,000 data). NonDemented persons exhibit no indications of dementia. VeryMildDemented has a few symptoms and is in the preliminary phase of the condition. Mild dementia presents notable yet still moderate symptoms of cognitive decline. Individuals with moderate dementia have more intense and intermediate signs of

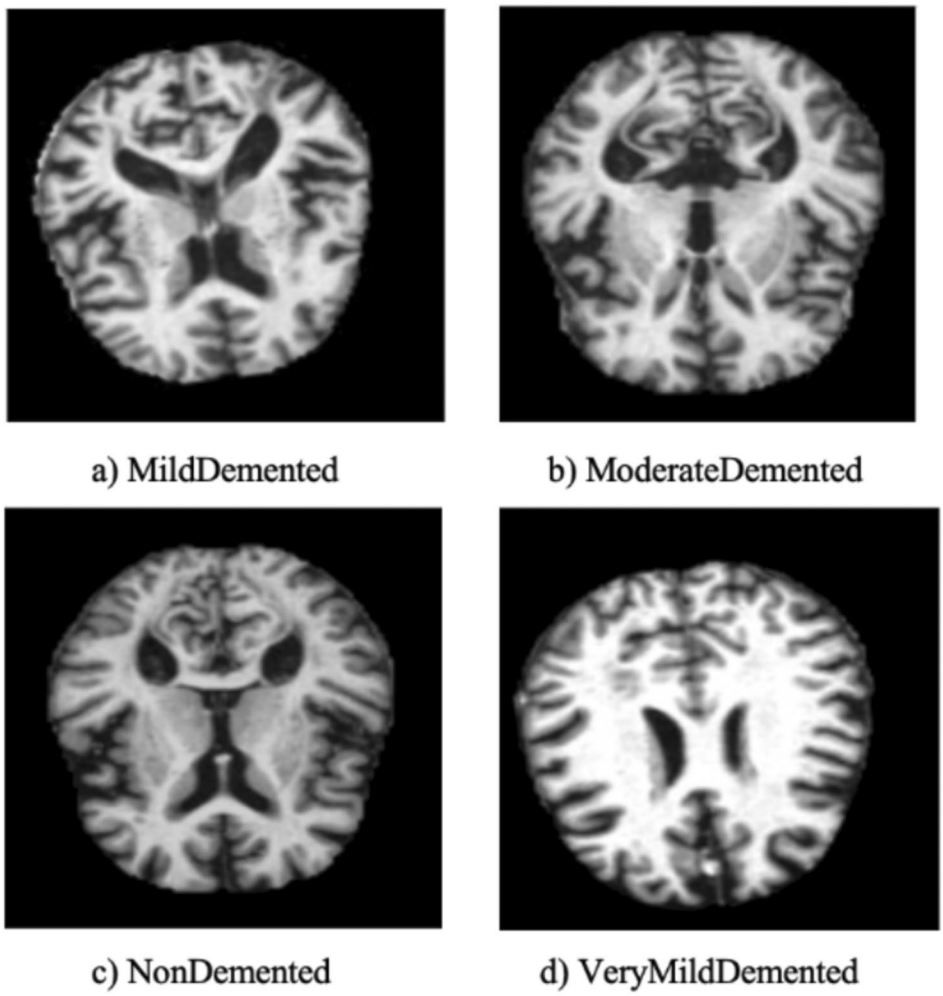

Fig. 1 Example MRI images from the normal group, showing typical brain structures used as a baseline for detecting different stages of Alzheimer's disease

the condition. All images in this collection are depicted as "skull-stripped," indicating that extraneous tissue and bone features outside the cranial region have been eliminated from the images. Exclusively incorporating brain tissue facilitates model concentration and mitigates the impact of extraneous information. The dimensions and resolution of the images have probably been adjusted to ensure they are sufficiently large for deep-learning models to analyze. Standardizing MRI images facilitates data processing and integration into the model.

Feature Extraction with Vision Transformers (ViTs) and EfficientNet

In recent years, deep learning-based feature extraction methods have significantly advanced medical image analysis and have become essential in disease detection. Feature extraction denotes the automated identification and extraction of salient and significant characteristics from photographs. This procedure is particularly crucial in the evaluation of MRI images utilized in diagnosing neurodegenerative disorders, including Alzheimer's disease. Deep learning models' feature extraction capabilities offer a more thorough and accurate analysis than older methods, facilitating more efficient identification of disease-specific visual biomarkers. In this context, EfficientNet and ViT architectures illustrate two different techniques for feature extraction.

The EfficientNet architecture is a novel methodology for deep learning, providing an innovative perspective on image classification and feature extraction tasks. This architecture enhances feature extraction efficiency compared to classic models by employing a compound scaling strategy that optimally balances the network's depth, width, and resolution. MBConv (Mobile Inverted Bottleneck Convolution) blocks, the fundamental component of EfficientNet, operate alongside pointwise convolution, depthwise convolution, and Squeeze-and-Excitation modules to effectively capture both low-level and high-level image characteristics. This feature extraction method is proficient in assessing intricate visual structures and identifying significant landmarks.

The feature extraction capabilities of EfficientNet arise from the architecture's novel topology. The composite scaling approach methodically adjusts the three essential dimensions of the model architecture (depth, width, and resolution), enhancing the efficacy of each dimension's contribution. In contrast with traditional convolutional neural networks (CNN), this method architecture preserves computational efficiency while augmenting model size and generating more effective feature representations. MBConv blocks effectively capture both local and global information due to their bottleneck architecture and inverted residual connections. Squeeze-and-Excitation modules provide channel-wise feature calibration, enabling the amplification of significant features while attenuating trivial ones. The amalgamation of these architectural elements renders EfficientNet an exceptionally proficient model for feature extraction.

Vision Transformers employ a transformer-based design instead of CNNs for computer vision tasks, where images are divided into small patches and processed sequentially through transformer layers [24]. A 224×224 pixel image is segmented into 196 patches of 16×16 pixels, and positional data are added to help the model understand the spatial relationships between patches [25].

The primary element of transformers is the multi-head self-attention mechanism [26]. This method acquires significant information by focusing on various areas of the image. These layers ascertain the interrelations among image segments and progressively refine the derived attributes.

The ViT models employed in the study vary in size, each optimized for a particular resolution and dimension (Fig. 2). The study employs two distinct ViT architectures: ViT Base 16/32 and ViT Large 16/32. ViT Base is one of the fundamental iterations within the ViT family. This model segments the input image into small blocks of 16×16 and 32×32 and processes each block through transformer layers. This model possesses limited parameters and is efficient for small to medium-sized datasets. It is mainly utilized in classification tasks that require high precision but do not demand extensive datasets. The ViT Large model is an expanded version of the model. It is optimized for extensive datasets and incorporates additional parameters.

Feature Selection Algorithms

Feature selection is an essential procedure for enhancing the efficacy of machine learning models and reducing computing complexity. This study employs three distinct feature selection methods to efficiently identify high-dimensional features extracted from brain MRI images and execute dimensionality reduction. ElasticNet (ENet), NCA, and MRMR algorithms each seek to uncover the most discriminative features in the dataset through distinct mathematical methodologies and optimization strategies. This section will study in detail the theoretical underpinnings, operational principles, and significance of feature selection algorithms in the context of Alzheimer's categorization.

Initially employed for feature selection, ENet is a robust linear regression model that combines Lasso (L1) and Ridge (L2) regressions. The mathematical representation of ENet appears as follows:

$$\min(1/(2n))\|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2{}^2 + \lambda_1\|\mathbf{w}\|_1 + \lambda_2\|\mathbf{w}\|_2$$

In this context, $\|w\|_1$ denotes the L1 norm penalty, $\|w\|_2{}^2$ signifies the L2 norm penalty, and λ_1 and λ_2 are the respective L1 and L2 regularization parameters. This

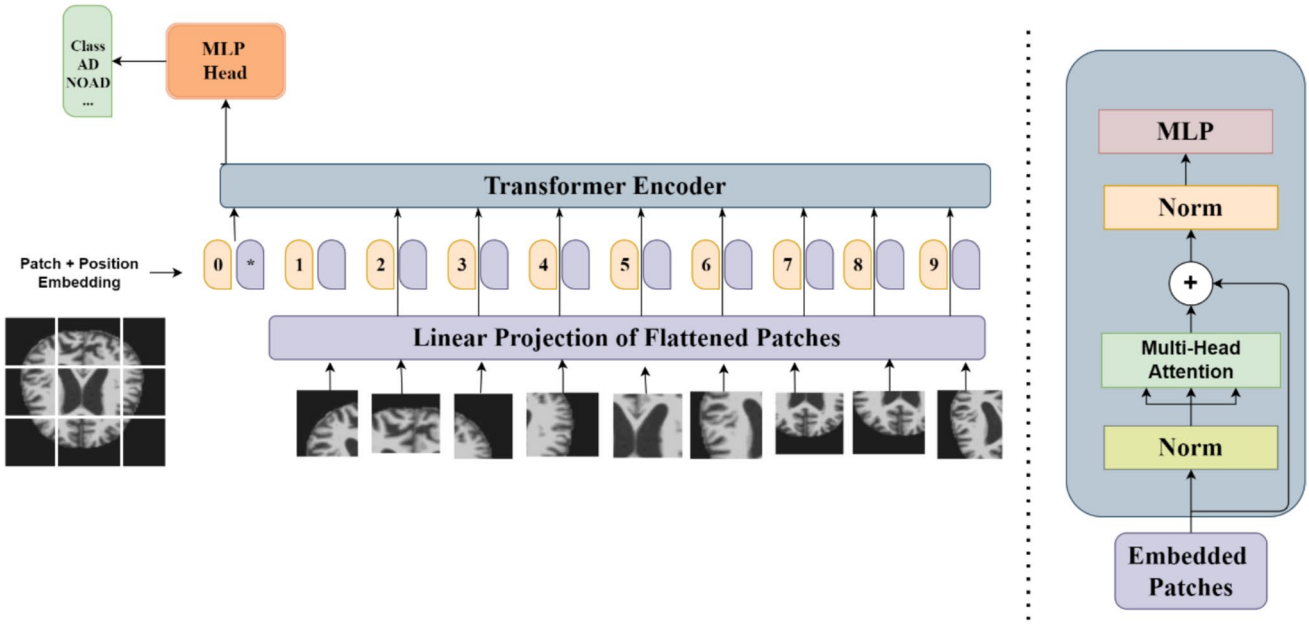

Fig. 2 The Vision Transformer (ViT) architecture illustrates the process of dividing MRI images into patches, embedding them, and extracting features through transformer layers

mathematical framework facilitates the selection of the most significant features from high-dimensional and associated data while simultaneously managing model complexity.

ENet is an efficient technique for diminishing the dimensionality of extensive feature sets, particularly from deep learning architectures like Vision Transformer. The L1 norm eliminates specific feature coefficients, whereas the L2 norm maintains a balance among highly linked features. This considerably decreases the amount of the dataset and optimizes processing time while retaining classification performance [27]. The characteristic of ENet renders it an optimal pick for feature identification, particularly in contexts involving high-dimensional datasets, such as medical image analysis.

NCA is a supervised dimensionality reduction technique that improves classification performance by refining the distance metrics among data items [28]. The algorithm's primary goal is to build a transformation matrix that maps data into a lower-dimensional space while preserving class associations [29].

In its mathematical framework, for a dataset, $X \in Rn \times d$, NCA learns a linear transformation matrix $A \in Rk \times d$ that maps data points from d-dimensional space to k-dimensional space through $yi = Axi$. The probability of point i choosing point j as a neighbor is defined as $pij \propto e^{\wedge}(-\|yi - yj\|^2)$. This transformation maximizes the likelihood of correct classification based on neighborhood relationships.

NCA's effectiveness stems from its ability to handle high-dimensional data while preserving essential class-specific features. It finds applications in various domains, including image recognition, natural language processing, and biological data analysis. The method enhances feature representation by bringing same-class data points closer while separating different classes [28, 29]. This approach particularly benefits distance-based classification methods by creating an optimal framework for accurate categorization, making it especially valuable for complex datasets like medical imaging analysis.

MRMR is a practical approach for feature selection grounded in two fundamental principles [30]. These concepts are characterized by maximum relevance and minimum redundancy. The mathematical representation of MRMR is articulated as follows:

$$\max\Phi(\mathbf{D}, \mathbf{R}), \Phi(\mathbf{D}, \mathbf{R}) = V(\mathbf{D}) - W(\mathbf{R})$$

In this context, V(D) quantifies the significance of features in relation to the target variable, whereas W(R) assesses the redundancy among the chosen features [31]. These values are typically computed via the mutual information metric:

$$V(\mathbf{D}) = (1/|\mathbf{S}|) \sum I(fi, c)$$
$$W(\mathbf{R}) = \left(1/|\mathbf{S}|^2\right) \sum I(fi, fj)$$

where I(fi,c) signifies the mutual information between feature fi and target class c, and I(fi, fj) denotes the mutual information between features.

MRMR is a proficient feature selection technique, particularly for high-dimensional datasets. The approach

enhances dataset size by picking features that exhibit high correlation with the target variable while maintaining low intercorrelation among themselves [30, 31]. This method enhances model efficacy, decreases training duration, and mitigates the likelihood of overfitting, particularly in high-dimensional data contexts like biological data analysis, picture processing, and genetic research.

Algorithms for Classifications

In this study, the performance of different machine learning algorithms for the classification of Alzheimer's disease is comparatively analyzed. Six distinct classification algorithms—support vector machine (SVM), gradient boosting machine (GBM), XGBoost, random forest (RF), logistic regression (LR), and k-nearest neighbors (KNN)—possess unique mathematical underpinnings and operational principles. These algorithms have been chosen for their widespread use in the literature, their adaptability to different data structures, and their reliable classification performance. To assess the efficacy of each algorithm, particularly in sensitive domains like medical picture analysis, the primary characteristics, mathematical principles, and classification methodologies of the algorithms will be scrutinized comprehensively.

SVM is a robust supervised learning method that identifies an ideal hyperplane to segregate data points with the maximum margin [32]. The mathematical representation of SVM is articulated as follows:

$$\min(1/2)\|\mathbf{w}\|^2 + C\Sigma\xi_i$$
$$\text{subject to: } y_i\left(\mathbf{w}\cdot\boldsymbol{x_i} + b\right) \geq 1 - \xi_i \text{ ve } \xi_i \geq 0$$

where w is the normal vector of the hyperplane, b is the shift term, C is the regularization parameter, and ξ_i is the slack variable. Kernel functions are used for nonlinear problems:

$$K\left(\mathbf{x_i}, \mathbf{X_j}\right) = \varphi\left(\mathbf{x_i}\right)\cdot\varphi\left(\mathbf{x_j}\right)$$

In this study, SVM was utilized to categorize high-dimensional brain MRI characteristics acquired from EfficientNet and ViT models [32]. Kernel functions' capacity to adeptly manage nonlinear data structures has enhanced the algorithm's efficacy, particularly in identifying various phases of Alzheimer's disease and categorizing nuanced morphological alterations in brain images. The margin maximization strategy of SVM enhanced class separation, yielding superior classification performance.

Random forest (RF) is a powerful classification technique that combines numerous decision trees with the notion of ensemble learning [33, 34]. The mathematical structure of RF is represented as follows:

$$\overset{\}}{f}_r\mathrm{f}(\mathbf{x}) = 1/B\sum_{i=1}\mathrm{B}f_i(\mathbf{x})$$

where B is the total number of trees and $f_i(x)$ is the prediction of each tree. Majority voting is used for classification:

$$\overset{\}}{y} = \mathrm{mode}\left\{f_i(\mathbf{x})\right\}_{i=1}B$$

RF samples data for each tree using a bootstrap aggregating (bagging) technique [35]:

$$X^* = \left\{\left(\mathrm{xi}_i^*, \mathrm{yi}_\mathrm{i}^*\right)\right\}_{\mathrm{i}=1}, \text{ where } \left(\mathrm{x}_\mathrm{i}^*, \mathrm{yi}_\mathrm{i}^*\right) \sim \overset{\}}{P}$$

This study demonstrated the successful utilization of RF to classify features derived from EfficientNet and ViT models. The primary advantage of the method is that each tree is trained on distinct data subsets and determines outcomes through majority voting [33–35]. This method prevents overlearning and ensures superior classification performance, even with imbalanced datasets. Integrating characteristics acquired from several views in medical image analysis allows the algorithm to yield dependable and uniform outcomes.

The other classifier model is the LR model. LR is a linear model created to address classification issues, extensively utilized in statistics and machine learning [36]. Logistic regression is employed to forecast the likelihood of an event transpiring and is especially effective in binary classification challenges. Notwithstanding the term "regression" in its designation, its objective is to categorize data and predict the class to which the data pertains rather than to estimate continuous values. Logistic regression is appropriate for categorization, in contrast to linear regression. Linear regression forecasts the dependent variable as a continuous value; nevertheless, this method is inappropriate for classification problems, as the dependent variable is typically categorical or binary. Logistic regression addresses this issue by constructing a model from linear combinations of independent variables and using a sigmoid function to convert the output of a linear equation into a constrained range (between 0 and 1) [37]. This function generates numbers that can be read as estimated probabilities, which are classified into categories based on a specific threshold value.

The sigmoid function is fundamental to logistic regression. The sigmoid function generates classification predictions by constraining the model's output within the range of 0 to 1 [38]. The sigmoid function is shown in Eq. 1.

$$\sigma(z) = \frac{1}{1 + e^{-z}} \cdots \tag{1}$$

In this equation, z is determined as a linear amalgamation of independent variables (e.g., $z = \mathrm{w0} + \mathrm{w1}\times 1 + \mathrm{w2}\times 2 + \ldots + \mathrm{w}n\times n$). The sigmoid function transforms this linear

combination into a number ranging from 0 to 1. If the output value exceeds 0.5, the model categorizes the input as one class (e.g., 1); if it is below, it categorizes it as another class (e.g., 0).

Logistic regression, while primarily utilized for binary classification, can also be employed for multiclass classification through methods such as one-vs-rest or softmax regression. In the one-vs-rest methodology, each class is forecasted in relation to the other classes using an individual binary classification model. Softmax regression, conversely, identifies the class with the highest probability by concurrently calculating the probabilities for each class in multiclass classifications. Logistic regression yields highly effective outcomes, particularly with linearly separable datasets. Logistic regression is both efficient and precise when the dataset does not necessitate intricate border delineations for accurate categorization. It is distinguished as an efficient classification model due to its lower processing demands compared to more intricate methods. Logistic regression may exhibit constrained efficacy on intricate and non-linear classification problems due to its reliance on linear bounds; yet, it serves as a robust model when linear precision is essential, establishing a decision boundary suitable for the data.

GBM is an effective machine learning system that minimizes error rates and enhances prediction accuracy by sequentially aggregating weak learners, typically decision trees. Using a boosting strategy, GBM adds a new model at each iteration by learning the faults of the preceding model [39, 40]. During this procedure, model parameters are refined by gradient descent, therefore minimizing the overall error [41].

The efficacy of GBM is contingent upon the precise calibration of parameters, including learning rate, tree quantity, and depth. Each additional tree incrementally enhances the model's accuracy by assimilating the residual mistakes from prior iterations. The algorithm proficiently comprehends intricate data patterns and methodically minimizes errors, particularly in high-dimensional datasets. Its proficiency in managing data noise and superior generalization capabilities render it an exceptionally beneficial option for intricate classification challenges.

XGBoost is an enhanced iteration of the gradient boosting method, tailored for challenges necessitating superior performance and accuracy [42]. Regularized learning and gradient descent techniques define the approach. It has features including L1 and L2 regularization, parallel processing capabilities, out-of-memory computation for extensive datasets, and management of missing data. Using advanced tree pruning mechanisms and growth algorithms, XGBoost [43] minimizes needless complexity while maintaining appropriate tree depth.

The algorithm's success is contingent upon precisely calibrating parameters like learning rate, maximum depth, number of trees, and sampling rate. It is distinguished by its capacity to proficiently identify data linkages and handle missing data, particularly in intricate datasets. These attributes render XGBoost a more advantageous choice compared to conventional boosting algorithms.

KNN is a supervised learning technique that classifies data points based on their proximity to neighboring ones [44]. The approach computes the distance of the data point to be classified from other instances in the dataset and predicts by assigning the majority class of the k-nearest neighbors. The efficacy of KNN is contingent upon the selection of k and the employed distance metric (Euclidean, Manhattan, or Minkowski). In high-dimensional areas, algorithm efficiency may diminish due to the "curse of dimensionality," which can be mitigated using dimensionality reduction techniques like PCA. Despite being a nonparametric and straightforward model, it is computationally demanding with large datasets. It is referred to as a "lazy learner" because it conducts categorization during prediction instead of learning during training. The efficacy of the model is contingent upon data preprocessing and the caliber of the training data, with performance enhancement achievable through approaches like normalization.

Proposed Method

This work presents the NCA-Enhanced Voting Algorithm for Alzheimer's Classification (NCA-EVA), which introduces a novel hybrid methodology that enhances feature selection and classification techniques (Fig. 3). This method is intended to effectively process and categorize features obtained from any feature extraction methodology and has two primary stages. Initially, the NCA method is employed to identify the most distinguishing features from the retrieved data. In the second stage, a Voting Classifier comprising KNN, SVM, and RF algorithms is utilized to categorize the chosen features. This method integrates the feature selection efficacy of NCA with the classification strength of the ensemble learning model to enhance classification performance and optimize processing duration.

The choice of algorithms employed in the NCA-EVA technique was predicated on the strengths and benefits of each algorithm. NCA is a feature selection approach tailored for classification tasks, utilized during the feature selection step. The primary rationale for selecting NCA is its capacity to learn the distance metric among features by allocating a weight to each feature. Thus, it can discern the characteristics that most effectively differentiate between classes and execute dimensionality reduction proficiently. Furthermore, in the experiments, the NCA algorithm achieved the most favorable outcomes among several feature selection algorithms, resulting in its preference for the proposed strategy.

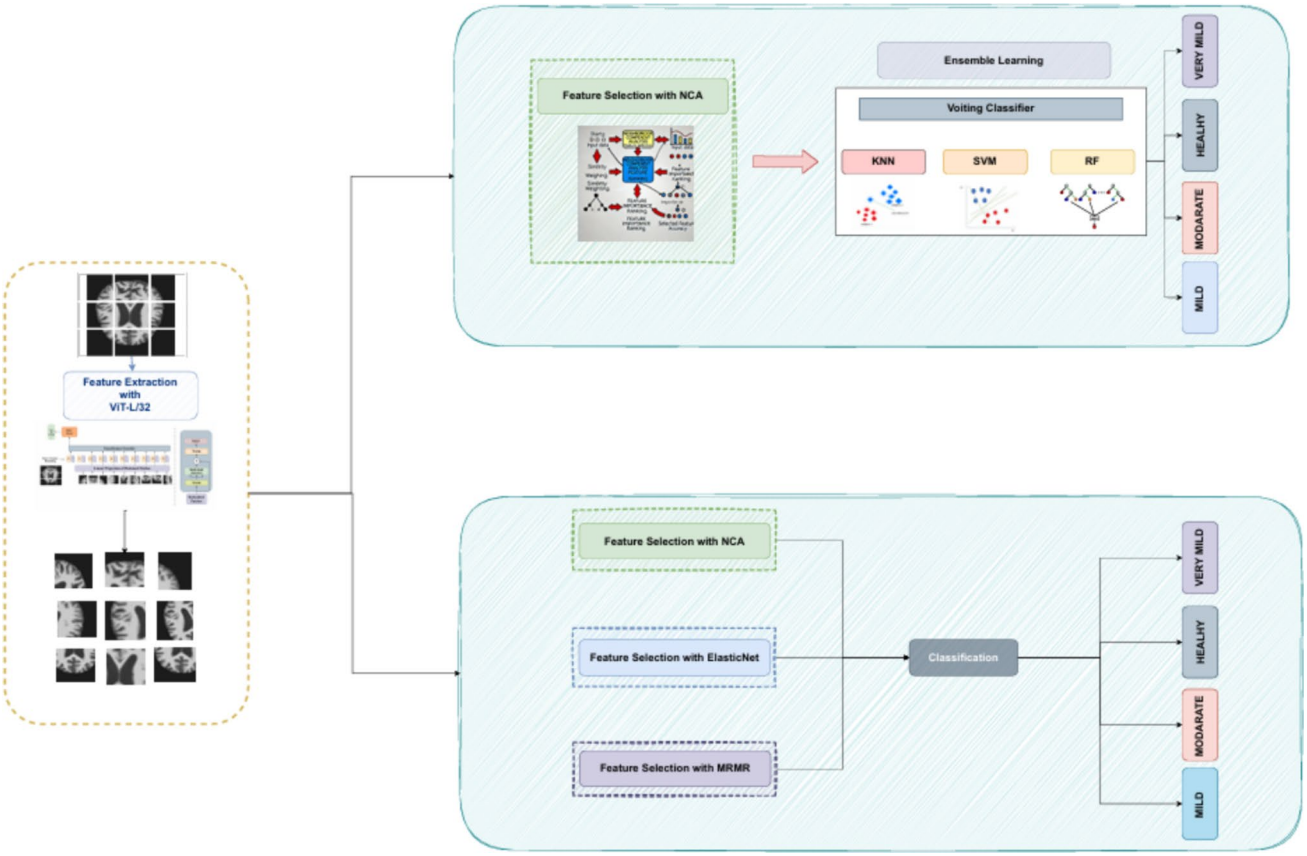

Fig. 3 Graphical abstract of the proposed NCA-EVA framework, showing the workflow from data preprocessing to feature extraction, feature selection, and final classification

Before model training, standard image preprocessing steps were applied to ensure consistency and improve model performance. All MRI images were resized to 224×224 pixels to match the input dimensions of the ViT and EfficientNet architectures. Pixel values were normalized to a [0,1] range, and data augmentation techniques such as horizontal and vertical flipping, random rotation, and slight brightness adjustments were applied to increase data diversity and reduce overfitting.

The ensemble model incorporates three distinct algorithms (KNN, SVM, and RF) selected to mitigate each other's limitations. The KNN algorithm is a straightforward and efficient classifier adept at identifying local patterns. The rationale for using the KNN algorithm in the proposed method is that it yields the most favorable outcomes in the experimental results. SVM excels in high-dimensional data and adeptly models intricate decision limits. Its resilience distinguishes RF from overfitting and its capacity to assess feature significance. The other two chosen algorithms, SVM and RF, also showed efficacy in experimental outcomes. The integration of these three algorithms with the Voting Classifier enhances classification performance by leveraging the characteristics of each method, resulting in more reliability and balance.

NCA, the first stage of the NCA-EVA method, is a metric learning approach for feature selection. NCA transforms features to maximize classification performance by modeling each instance's relationship with other instances in a probabilistic framework. Using the Euclidean distance between instances, the approach learns a transformation matrix such that instances in the same class are close to each other and instances in different classes are far apart.

The basic equation of NCA is as follows:

$$p(\mathrm{i} \rightarrow \mathrm{j}) = \exp\left(-\|\mathrm{Wx_i} - \mathrm{Wx_j}\|^2\right) / \Sigma_\{k \neq \mathrm{i}\} \exp\left(-||\mathrm{Wx_i} - \mathrm{Wx_k}\|^2\right)$$

In this equation,

- $p(\mathrm{i} \rightarrow \mathrm{j})$: the probability that instance i select instance j as a neighbor
- W: learned transformation matrix
- x_i and x_j: sample pairs

- $\|Wx_i - Wx_j\|^2$: square of the Euclidean distance between samples in transformed space
- $\Sigma_\{k \neq i\}$: sum over all samples different from i
- exp: exponential function

This probability function is optimized to maximize the probability that instances in the same class are close to each other. An optimal transformation in the feature space is obtained by updating the W matrix in an iterative process.

Min–Max normalization is utilized during the second phase of the NCA-EVA methodology to standardize the chosen characteristics. The normalization procedure seeks to enhance classifier performance by transforming features of varying scales into a uniform scale within the range [0,1]. Min–Max normalization executes a linear transformation utilizing the minimum and maximum values of each attribute.

The fundamental equation of Min–Max normalization:

$$x'_i = (x_i - x_min)/(x_max - x_min)$$

$x'_i = (x_i - x_min)/(x_max - x_min)$.

In this equation,

- x'_i: normalized feature value
- x_i: original feature value
- x_min: minimum value of the property
- x_max: maximum value of the property

As a result of this normalization process:

- All features are scaled in the range [0,1].
- The structure of the original distribution is preserved.
- Reduced impact of outliers.
- Features at different scales become comparable.

The concluding phase of the NCA-EVA approach, Ensemble Voting, is a soft voting technique that amalgamates the predictions of three distinct classifier algorithms: KNN, SVM, and RF. This method involves each classifier generating a probability estimate for a class, with the final choice derived from the average of these probabilities. The soft voting technique considers the predictive confidence of each classifier, yielding a more dependable categorization.

The fundamental equation of Ensemble Voting classification:

$$P(y \mid x) = (1/M)\Sigma_[m = 1]^{\wedge} M P_m(y \mid x)$$

In this equation,

- $P(y|x)$: final probability of class y for instance x
- M: total number of classifiers ($M = 3$)
- P_m(y|x): m. classifier's probability estimate of class y for sample x
- Σ: sum over all classifiers

Final classification decision:

$$\hat{y} = argmax_y\ P(y|x)$$

where:

- $\hat{y}$: predicted class label
- argmax_y: operator that selects the class label with the highest probability
- $P(y|x)$: ensemble probability calculated for each class

This voting mechanism leverages the strengths of each classifier, resulting in a more robust and reliable classification performance.

This ensemble model was implemented using soft voting. Soft voting calculates the probability of each model belonging to a class and predicts the class with the highest probability. Thus, the strong classification features of each model are exploited, and errors are minimized. To improve this method's overall performance, prediction with soft voting, which considers the information provided by each model in classification, provides more reliable results in a serious classification problem such as Alzheimer's disease. The process of multiclass classification using feature extraction and a voting ensemble model is presented as pseudo-code in Table 2.

Experimental Results

This paper presents the results of several experiments and the proposed NCA-EVA method to improve the classification performance of Alzheimer's disease. The studies are discussed in two parts. In the first part, 66 models are trained on four-class data. Then, the NCA-EVA method is applied to 6 models with two-class data, and results are obtained for comparison with two-class studies in the literature. To ensure the robustness and reliability of the proposed method, a fivefold cross-validation strategy was applied. The dataset was randomly partitioned into five equal subsets, where four folds were used for training and one fold for testing in each iteration. This process was repeated five times, and the final performance metrics were reported as the average of these iterations. The dataset used in this study is balanced across the four classes, thereby minimizing the risk of class imbalance. Additionally, the use of cross-validation helped reduce the risk of overfitting and ensured that the reported results accurately reflect the generalization performance of the model.

Table 2 Pseudo-code of the proposed method

```
procedure EXTRACT_FEATURES(data_path, model_path)
    model ← LOAD_MODEL(model_path)
    data ← LOAD_DATA(data_path)
    features ← [model.EXTRACT(sample) for sample in data]
    return features
end procedure
procedure TRAIN_ENSEMBLE_MODEL(features, labels)
    train_data, val_data, test_data ← SPLIT_DATA(features, labels)
    reduced_features ← APPLY_NCA(train_data, val_data, test_data, n_components=300)
    normalized_features ← APPLY_MINMAX_SCALER(reduced_features)
    knn ← INITIALIZE_KNN()
    svm ← INITIALIZE_SVM()
    rf ← INITIALIZE_RF()
    ensemble_model ← CREATE_VOTING_MODEL([knn, svm, rf], voting_type='soft')
    ensemble_model.TRAIN(normalized_features.train, train_data.labels)
    predictions ← ensemble_model.PREDICT(normalized_features.test)
    metrics ← EVALUATE(predictions, test_data.labels)
    return metrics
end procedure
```

Table 3 Traditional classification methods: accuracy and processing time comparison

	Accuracy (%)	Time (s)
EfficientNet_V2_L	78.37	18,867
EfficientNet_B7	81%	17,261
ViT-B/16	62.11	6048
ViT-B/32	40.43	1668
ViT-L/16	63.72	35,100
ViT-L/32	90.25	3780

First, the traditional classification process was performed with the most complex models of EfficientNet architecture (EfficientNet_B7 and EfficientNet_v2_L) and ViT models, and the results are given in Table 3.

Table 3 compares the performance of various deep learning models in classifying Alzheimer's disease. The ViT-L/32 model attains its highest performance, with an accuracy of 90.25% in 3780 s. The EfficientNet models (B7 and V2/L) had commendable accuracy rates of 81% and 78.37%, respectively, although they necessitated significantly longer processing times of 17,261 and 18,867 s. ViT models generally yield results more rapidly; nevertheless, the low accuracy of 40.43% for ViT-B/32 suggests that both processing time and the accuracy-duration trade-off must be evaluated in model selection.

This study observed that the most successful result was obtained with the ViT-L/32 model when looking at traditional classification models. For this reason, this model was used for feature extraction in the rest of the study, and the results are given in Table 4.

Table 4 compares the performance of various classification methods utilizing characteristics derived from the ViT-L/32 model. The KNN method had the highest performance, with an accuracy of 89.01%, alongside precision, recall, and F1-score of 89%, and was the most rapid model, requiring merely 2.81 s for processing. The XGBoost algorithm achieved the second-highest accuracy at 87.74%, followed by SVM at 83.94%, RF at 83.48%, and LR at 82.17%, demonstrating commendable performance. The GBM algorithm had the lowest performance, achieving an accuracy of 73.65%. Regarding processing time, SVM is the slowest model at 2243.57 s, while KNN is notably rapid. The findings indicate that the features derived from ViT-L/32 yield effective outcomes for both accuracy and speed, mainly when utilized with algorithms like KNN and XGBoost.

Table 4 Effect of NCA-selected ViT-L/32 features on machine learning models: accuracy and time evaluation

	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Time (s)
SVM	83.94	84	84	84	2243.57
KNN	89.01	89	89	89	2.81
RF	83.48	83	83	83	168.62
LR	82.17	82	82	82	59.44
GBM	73.65	73	74	73	100.42
XGBoost	87.74	88	88	88	97.14

As seen in Table 4, the most successful result is obtained with the feature extraction result of the ViT-L/32 model. For this reason, in the rest of the study, the features obtained with the ViT-L/32 model were selected as the most meaningful ones with different feature selection algorithms, and the classification process was performed. First, the most significant features extracted with the ViT-L/32 model were selected with NCA, and classification was performed using different classification algorithms.

The outcomes from implementing NCA feature selection demonstrate considerable enhancements relative to the preceding table (Table 5). The KNN method exhibits the most significant enhancement, with its accuracy rate rising from 89.01 to 97.86%, and it remains the swiftest model, requiring 3.41 s for processing. The XGBoost algorithm demonstrated a notable enhancement, with an accuracy of 91.72% and attaining 92% across all measures (precision, recall, F1-score). The RF algorithm had the third-highest accuracy at 88.19%, whereas the other algorithms exhibited overall enhancements in performance. Only the processing duration of the GBM algorithm significantly increased to 5869.45 s. The results indicate that the NCA feature selection method enhances the quality of features derived from ViT-L/32 and significantly improves classification performance. In the rest of the study, the most significant features extracted with the ViT-L/32 model were selected with ENet and then classified with different classification algorithms.

Looking at the results obtained after using the ENet feature selection method in Table 6, the KNN algorithm showed the best performance with an accuracy rate of 90.45% and 90% values in all metrics. It worked very fast, with a processing time of 10.31 s. The XGBoost algorithm performed second best with an accuracy of 87.57%, while the RF algorithm ranked third with 83.84%. The GBM algorithm performed the worst, with an accuracy of 43.25% and unbalanced precision-recall values (41% and 55%), and was the slowest model, with a processing time of 1104.60 s. These results show that the ENet feature selection method performs poorly compared to NCA, especially in the KNN algorithm, where there is a decrease of about 7% compared to the accuracy of 97.86% obtained with NCA.

Finally, the most significant features extracted with the ViT-L/32 model were selected with MRMR and then classified with different classification algorithms (Table 7).

As seen in Table 7, after using the MRMR feature selection method, the KNN algorithm performed the best with an

Table 5 Effect of NCA-selected ViT-L/32 features on machine learning models: accuracy and time evaluation

	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Time (s)
SVM	83.65	84	84	84	1723.03
KNN	97.86	98	98	98	3.41
RF	88.19	88	88	88	49.40
LR	82.25	82	82	82	45.09
GBM	78.07	78	78	78	5869.45
XGBoost	91.72	92	92	92	77.11

Table 6 Effect of ENet-selected ViT-L/32 features on machine learning models: accuracy and time evaluation

	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Time (s)
SVM	82.08	82	83	82	536.99
KNN	90.45	90	90	90	10.31
RF	83.84	84	84	84	146.28
LR	80.67	81	81	81	35.73
GBM	43.25	41	55	35	1104.60
XGBoost	87.57	88	88	88	52.50

Table 7 Effect of MRMR-selected ViT-L/32 features on machine learning models: accuracy and time evaluation

	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Time (s)
SVM	75.68	76	76	76%	588.04
KNN	88	88	88	88%	1.35
RF	83.48	83	83	83%	358.42
LR	82.31	82	82	82%	236.45
GBM	71.75	71	72	71%	2050.64
XGBoost	85.36	85	85	85%	188.08

accuracy of 88% and consistent values of 88% in all metrics (precision, recall, F1-score). It was the fastest model, with a processing time of 1.35 s. The XGBoost algorithm performed second best with an accuracy of 85.36%, while the RF algorithm ranked third with 83.48%. GBM and SVM algorithms demonstrated lower performance, with accuracy rates of 71.75% and 75.68%, respectively. Notably, GBM, the slowest model, required a processing time of 2050.64 s. These results show that the MRMR feature selection method performs poorly compared to NCA and ENet methods, especially in the KNN algorithm, which gives the best result compared to the accuracy rates of 97.86% obtained with NCA and 90.45% obtained with ENet.

These studies show that the most successful result is obtained when feature selection is performed with NCA. After this step, the performance of different feature extraction and classification models of the NCA feature selection model is analyzed and compared to the proposed NCA-EVA model. First, the ViT-B/16 model was used for feature extraction (Table 8).

As shown in Table 8, in the comparative analysis of the features extracted from the ViT-B/16 model, the proposed NCA-EVA method showed the highest performance with an accuracy of 98.22% and 98% in all metrics. It worked very fast, with a processing time of only 4.69 s. The KNN algorithm performed second best with an accuracy of 97.58%, while XGBoost ranked third with 91.86%. RF (86%), SVM (82.60%), and LR (82.12%) algorithms performed moderately, while the GBM algorithm performed the worst with an accuracy of 77.32% and was the slowest model with a processing time of 6348.82 s. These results clearly show that the proposed NCA-EVA method outperforms other classification algorithms in terms of both accuracy and processing time.

In the rest of the study, the ViT-B/32 model was used for feature extraction (Table 9). In the comparative analysis of the features extracted from the ViT-B/32 model, the proposed NCA-EVA method performed the best with an accuracy of 97.78% and 98% in all metrics. It was the fastest model, with a processing time of only 1.85 s. The KNN algorithm performed the second best with an accuracy of 96.7%, while RF and XGBoost algorithms ranked third and fourth with close accuracy of 88.52% and 88.33%, respectively. SVM (75.81%), LR (80.15%), and GBM (74.05%) algorithms showed lower performance, especially GBM, which was the slowest model with a processing time of 1905.14 s. These results show that the NCA-EVA method outperforms other classification algorithms in terms of both accuracy and processing time when feature extraction is performed with the ViT-B/32 model.

In the rest of the study, the ViT-L/16 model was used for feature extraction (Table 10). In the comparative analysis of the features extracted from the ViT-L/16 model, the proposed NCA-EVA method performed the best with an accuracy of 98.05% and 98% in all metrics, achieving a speedy processing time of 5.93 s. The KNN algorithm performed second best with an accuracy of 96.64%, followed by XGBoost with 89.32% and RF with 86.70%. The LR

Table 8 Comparison of ViT-B/16 feature extraction and NCA feature selection across classifiers with results of the proposed NCA-EVA method

	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Time (s)
SVM	82.60	83	83	83	1895.54
KNN	97.58	98	98	98	26.68
RF	86	86	86	86	85.06
LR	82.12	82	82	82	44.54
GBM	77.32	77	77	77	6348.82
XGBoost	91.86	92	92	92	58.62
NCA-EVA (proposed method)	98.22	98	98	98	4.69

Table 9 Comparison of ViT-B/32 feature extraction and NCA feature selection across classifiers with results of the proposed NCA-EVA method

	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Time (s)
SVM	75.81	76	76	76	1005.41
KNN	96.7	97	97	97	28
RF	88.52	89	89	88	159.14
LR	80.15	80	80	80	46.03
GBM	74.05	74	74	74	1905.14
XGBoost	88.33	88	88	88	15.82
NCA-EVA (proposed method)	97.78	98	98	98	1.85

(81.52%), SVM (76.70%), and GBM (75.19%) algorithms performed worse, especially GBM, which was the slowest model with a processing time of 1911.53 s. These results show that the NCA-EVA method outperforms the other classification algorithms in terms of accuracy and processing time when feature extraction is performed with the ViT-L/16 model. This model gives consistent results with other ViT models.

In the rest of the study, the ViT-L/32 model was used for feature extraction. As shown in Table 11, in the comparative analysis of the features extracted from the ViT-L/32 model, the proposed NCA-EVA method showed the highest performance with an accuracy of 98.97% and 99% in all metrics. It was the fastest model, with a processing time of 1.26 s. The KNN algorithm performed second best with an accuracy of 97.33% and metric values of 97%. XGBoost and RF performed well with 91.72% and 88.19%, respectively, while SVM (83.65%), LR (82.25%), and GBM (78.07%) algorithms achieved lower accuracy rates. The GBM algorithm was the slowest model, with a processing time of 5869.45 s. These results clearly show that the NCA-EVA method provides a significant advantage over all other classification algorithms in terms of accuracy and processing time when used for feature extraction of the ViT-L/32 model.

In another study phase, the EfficientNet_B7 model was used for feature extraction. As shown in Table 12, in the comparative analysis of the features extracted from the EfficientNet_B7 model, this time, the KNN algorithm performed the best with an accuracy of 98.55% and 99% on all metrics. The proposed NCA-EVA method performed second best with an accuracy of 97.86% and 98% on all metrics, but with a processing time of 3.41 s, much faster than KNN's 25.57 s. XGBoost and RF performed well with 91.10% and 89.20%, respectively, while SVM (75.41%), LR (74.41%),

Table 10 Comparison of ViT-L/16 feature extraction and NCA feature selection across classifiers with results of the proposed NCA-EVA method

	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Time (s)
SVM	76.70	77	77	77	872.48
KNN	96.64	97	97	97	25.15
RF	86.70	87	87	87	48.28
LR	81.52	81	82	81	46.70
GBM	75.19	75	75	75	1911.53
XGBoost	89.32	89	89	89	16.95
NCA-EVA (proposed method)	98.05	98	98	98	5.93

Table 11 Comparison of ViT-L/32 feature extraction and NCA feature selection across classifiers with results of the proposed NCA-EVA method

	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Time (s)
SVM	83.65	84	84	84	1723.03
KNN	97.33	97	97	97	25.02
RF	88.19	88	88	88	49.40
LR	82.25	82	82	82	45.09
GBM	78.07	78	78	78	5869.45
XGBoost	91.72	92	92	92	77.11
NCA-EVA (proposed method)	98.97	99	99	99	1.26

Table 12 Comparison of EfficientNet_B7 feature extraction and NCA feature selection across classifiers with results of the proposed NCA-EVA method

	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Time (s)
SVM	75.41	75	75	75	1405.24
KNN	98.55	99	99	99	25.57
RF	89.20	89	89	89	78.17
LR	74.41	74	74	74	14.91
GBM	73.92	73	74	74	1903.89
XGBoost	91.10	91	91	91	16.25
NCA-EVA (proposed method)	97.86	98	98	98	3.41

and GBM (73.92%) algorithms achieved lower accuracy rates. The GBM algorithm was the slowest model, with a processing time of 1903.89 s. These results show that the KNN and NCA-EVA methods are much more successful than other classification algorithms when used in the feature extraction of the EfficientNet_B7 model. At the same time, NCA-EVA provides a significant advantage in terms of speed.

Finally, the EfficientNet_v2_L model is used for feature extraction. Looking at Table 13, the comparative analysis of the features extracted from the EfficientNet_v2_L model shows a significant decrease in the performance of all models. The NCA-EVA method demonstrated the highest performance, achieving an accuracy of 42.5% and 42% across all metrics, and was one of the fastest models, with a processing time of 8.23 s. The RF algorithm performed close to NCA-EVA with 42.82%, while XGBoost was the third best with 41.81%. KNN (39.76%), GBM (38.64%), SVM (37.88%), and LR (37.58%) algorithms showed meager accuracy rates. The GBM algorithm was again the slowest model, with a processing time of 1920.62 s. These results clearly show that the EfficientNet_v2_L model is unsuitable for feature extraction and performs much worse than the previous models.

When the studies in the literature are examined, data are generally divided into two classes. For this reason, the proposed NCA-EVA method was finally used on two-class Alzheimer's data, and the results are given in Table 14.

According to the results presented in Table 14, the NCA-EVA method was tested on two classes of Alzheimer's data with different feature extraction models. The EfficientNet_B7 model performed the best with an accuracy of 99.93% and very high values in all metrics (precision: 99.95%, recall: 99.90%, F1-score: 99.93%). ViT-B/16 and ViT-L/32 performed second best with 99.87% accuracy, while ViT-L/16 and ViT-B/32 performed similarly well with 99.85% and 99.67%, respectively. The EfficientNet_v2_L model, on the other hand, performed significantly worse than the other models, with an accuracy of 72.57%. Regarding processing time, all models (except EfficientNet_B7) produced speedy results, running in under 1 s. These results show that the proposed NCA-EVA method performs well in two-class Alzheimer's classification, especially with EfficientNet_B7 and ViT models.

When the results of all experimental studies are evaluated, it is seen that the proposed NCA-EVA method achieves the best performance with the ViT-L/32 model. In this combination, the NCA-EVA method achieved 98.97% accuracy, 99% precision, recall, and F1-score values, and a processing time of only 1.26 s. In tests with different feature extraction models (ViT-B/16, ViT-B/32, ViT-L/16, EfficientNet_B7), NCA-EVA consistently achieved accuracy rates in the range of 97–98%, outperforming traditional machine learning algorithms (SVM, KNN, RF, LR, GBM, XGBoost) in terms of both accuracy and processing time. Especially in computation time, NCA-EVA runs in the range of 1–8 s compared to 1900–6000 s for GBM, demonstrating the computational efficiency of the method. When only the EfficientNet_v2_L model was used, the performance of all algorithms decreased significantly (around 42%), showing the importance of feature extraction model selection.

Table 13 Comparison of EfficientNet_v2_L feature extraction and NCA feature selection across classifiers with results of the proposed NCA-EVA method

	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Time (s)
SVM	37.88	36	38	36	14,504.73
KNN	39.76	40	40	39	10.12
RF	42.82	42	43	42	82.81
LR	37.58	36	38	36	14.80
GBM	38.64	38	39	37	1920.62
XGBoost	41.81	41	42	41	52.44
NCA-EVA (proposed method)	42.5	42	42	42	8.23

Table 14 Results of classification of different feature extraction models with NCA-EVA

	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Time (s)
EfficientNet_B7	99.93	99.95	99.90	99.93	0.85
EfficientNet_v2_L	72.57	73	73	73	6.23
ViT-B/16	99.87	99.7	99.99	99.85	0.84
ViT-B/32	99.67	99.95	99.31	99.63	0.83
ViT-L/16	99.85	99.99	99.65	99.83	0.79
ViT-L/32	99.87	99.9	99.8	99.85	0.74

Conclusion and Discussion

In this study, we proposed NCA-EVA, an approach designed to classify different stages of Alzheimer's disease with high accuracy and exceptionally short processing time. The method effectively addresses several key limitations of existing approaches in multi-stage Alzheimer classification. Using a four-class dataset, our model achieved an accuracy of 98.97% in distinguishing No Dementia, Very Mild Dementia, Mild Dementia, and Moderate Dementia. When reduced to a two-class problem, the accuracy further increased to 99.87%, demonstrating the robustness and adaptability of the proposed framework. Moreover, NCA-EVA achieved these results with a processing time that was approximately 1200 times faster than traditional NCA-based methods, completing inference in just 1.26 s. This computational efficiency highlights its potential for real-time clinical deployment as a decision support system.

When compared with previous studies, our results represent a significant advancement. For instance, Sasikala et al. [15] reported an accuracy of 99.26% for a binary classification problem using a deep CNN. In contrast, our method achieved 99.87%, offering a slight but consistent improvement with far greater processing speed. Other ViT-based studies, such as Xin et al. [18] and Soundarya et al. [19], reported accuracy rates in the range of 90–96% for multi-class Alzheimer classification. In contrast, our four-class model reached 98.97%, surpassing these benchmarks. This demonstrates that combining advanced feature extraction techniques with the hybrid NCA-EVA strategy provides a clear performance benefit over single-model approaches. The improvement is not only numerical but also methodological, as our method integrates the strengths of multiple classifiers while drastically reducing computational costs.

The clinical relevance of these findings is substantial. Alzheimer's disease diagnosis is often time-consuming and relies heavily on expert interpretation of MRI scans. The ability of NCA-EVA to deliver near-instant, highly accurate predictions can support neurologists and neuroradiologists by reducing diagnostic workload and enabling earlier interventions. For example, in busy hospital environments, the model could serve as a real-time triage tool, quickly identifying cases requiring urgent follow-up. Furthermore, the extremely short runtime ensures that the method can be integrated into clinical workflows without adding delays, which is particularly important for screening programs and resource-limited healthcare settings. These advantages highlight the translational potential of our work from computational research to real-world medical applications.

Despite these promising results, several limitations should be acknowledged. First, the dataset used in this study consists of 44,000 MRI slices, which represent multiple images from a smaller group of patients rather than 44,000 unique individuals. Although the data provider prepared the dataset in a pre-balanced and pre-shuffled format to minimize overlap, there remains a possibility of correlation between slices from the same patient. This could introduce bias if images from one patient were inadvertently distributed across training, validation, and test sets. While this risk is expected to have minimal effect due to the balanced nature of the dataset and the comparative focus of this study, it remains a methodological limitation. Future studies will explicitly apply patient-level splitting to ensure generalizability further and reduce potential bias.

Second, this study did not utilize a real-world, independent external dataset for validation. While the proposed method demonstrates exceptional performance on the current dataset, independent validation using diverse, real-world clinical data is essential before clinical deployment. Additionally, no direct collaboration with clinicians or neuroradiologists was conducted during this study, which will be addressed in future work by incorporating expert feedback and clinical test scenarios.

In conclusion, the NCA-EVA method emerges as a highly accurate, efficient, and clinically relevant solution for the early detection and staging of Alzheimer's disease. Compared to conventional machine learning algorithms and existing deep learning models, it provides a substantial reduction in processing time and sets a new benchmark for multi-class Alzheimer classification. By addressing the limitations noted above—particularly through patient-level data handling and external validation—this approach has the potential to be integrated into clinical diagnostic pipelines. Beyond Alzheimer's disease, the underlying framework can be extended to other neurodegenerative conditions and complex medical imaging challenges, advancing the field of AI-driven healthcare diagnostics.

Data Availability The data supporting this study's findings are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Declarations

Conflict of Interest The authors declare no competing interests.

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